

Name _____

Section _____

MTH:2310 DISCRETE MATH
MIDTERM 2

Complete all of the following problems being as detailed as you can be. Remember to fully define new variables/symbols. Even if you cannot completely solve a problem, a good description of key terms and concepts relevant to the problem is sufficient for some partial credit. Do your best to make your solutions organized and written legibly.

Handwriting. (4 points)

Your ability to express your solutions is just as important as mathematical accuracy.

You will be penalized up to 2 points for arithmetic or notation errors.

You will be penalized up to 2 points for disorganized or hard-to-read solutions.

Problem 1. (8 points) List all of the elements in the following set.

$$\{x \mid (x = 2n + 1) \wedge (n \in \mathbb{N}) \wedge (81 - x^2 > 0)\}$$

Problem 2. (6 points) Determine whether the following statements are true or false.

(a) (2 points) $\left| \left\{ \{3, 2, 3\}, \emptyset, 1, \{2, 3, 2\}, \{\emptyset\}, 3 \right\} \right| = 5$

(b) (2 points) $\left\{ \left\{ \{\emptyset\} \right\} \right\} \subseteq \left\{ \{3, 2, 3\}, \emptyset, 1, \{2, 3, 2\}, \{\emptyset\}, 3 \right\}$

(c) (2 points) $\{2, 3\} \in \left\{ \{3, 2, 3\}, \emptyset, 1, \{2, 3, 2\}, \{\emptyset\}, 3 \right\}$

Problem 3. (14 points)

- (a) (8 points) Identify the argument form for the following argument. Fully define all introduced variables.

"If the for-loop is faulty or the while-loop is faulty, then the hardware is unreliable or the output is incorrect."

"The for-loop is faulty."

"The hardware is not unreliable. "

"Therefore, the output is incorrect."

- (b) (6 points) Determine whether the argument in part (a) is valid or invalid. Justify your answer by referencing appropriate rules of inference, or by using the definition of a valid argument.

Name _____

Section _____

Problem 4. (8 points) What rule of inference is being used in each of the following arguments?

(a) (4 points) *"Steve went to the gym today, or he stayed home. Steve left the house today. Therefore, Steve went to the gym today."*

(b) (4 points) *"For an integer n , the number $(n^3 - n)$ is even. Therefore, the number $((-17)^2 - (-17))$ is even."*

Problem 5. (8 points) Let A , and B be non-empty sets. Let U be the universal set. Show that $\overline{A \setminus B} = \overline{A} \cup B$ using set-builder notation only.

Problem 6. (8 points) Use a direct proof to prove that if x and y are rational numbers, then $x + y$ is also a rational number.

Problem 7. (8 points) Use a proof by contraposition to prove that if x^2 is irrational, then x is irrational.

Name _____

Section _____

Problem 8. (28 points) Let A, B, C, D be sets and U be the universal set. Suppose the sets satisfy the following relationships.

- $A \subseteq B$
- $C \subseteq B$
- $D \subseteq B$
- $C \cap D = \emptyset$
- $A \cap C \neq \emptyset$
- $A \cap D \neq \emptyset$
- $C \setminus A \neq \emptyset$
- $D \setminus A \neq \emptyset$

(a) (16 points) Use a Venn diagram to illustrate all of these relationships.

(b) (6 points) Redraw the diagram and shade the region corresponding to

$$\overline{B} \cup (C \cap A) \cup (D \setminus A)$$

Problem 9. (8 points) Let $f : \mathbb{N} \rightarrow \mathbb{N}$ be the function defined by $f(n) = n^2 + 1$. Determine whether $f(n)$ is injective, surjective, both, or neither. You do not need to prove your answer, but you must justify it.

Problem 10. (8 points) Let $A = \{1, 2, 3\}$, $B = \{w, x, y, z\}$, $C = \{\alpha, \beta, \gamma\}$. Define the functions $f : B \rightarrow C$ and $g : A \rightarrow B$ explicitly or with a “bubble-arrow” diagram such that their composition, $f \circ g : A \rightarrow C$ is bijective

Name _____

Section _____

Modus Ponens	$((p \Rightarrow q) \wedge q) \Rightarrow q$
Modus Tollens	$((p \Rightarrow q) \wedge (\sim q)) \Rightarrow \sim p$
Hypothetical Syllogism	$((p \Rightarrow q) \wedge (q \Rightarrow r)) \Rightarrow (p \Rightarrow r)$
Disjunctive Syllogism	$((p \vee q) \wedge (\sim p)) \Rightarrow q$
Addition	$p \Rightarrow (p \vee q)$
Simplification	$(p \wedge q) \Rightarrow p$
Conjunction	$((p) \wedge (q)) \Rightarrow (p \wedge q)$
Resolution	$((p \vee q) \wedge (\sim p \vee r)) \Rightarrow (q \vee r)$
Universal Instantiation	$\forall x P(x) \Rightarrow P(c)$
Universal Generalization	$P(c) \Rightarrow \forall x P(x)$
Existential Instantiation	$\exists x P(x) \Rightarrow P(c)$
Existential Generalization	$P(c) \Rightarrow \exists x P(x)$
Universal Modus Ponens	$(\forall x (P(x) \Rightarrow Q(x)) \wedge P(c)) \Rightarrow Q(c)$
Universal Modus Tollens	$(\forall x (P(x) \Rightarrow Q(x)) \wedge (\sim Q(c))) \Rightarrow \sim P(c)$